Earth and Planetary Science Letters, 117 (1993) 29–42
Elsevier Science Publishers B.V., Amsterdam

[UC]

Clockwise rotation of the Red River fault inferred from paleomagnetic study of Cretaceous rocks in the Shan–Thai–Malay block of western Yunnan, China

Shoubu Funahara [a,†], Nobukazu Nishiwaki [a], Fumiyuki Murata [b], Yo-ichiro Otofuji [a] and Yi Zhao Wang [c]

[a] *Department of Earth Sciences, Faculty of Science, Kobe University, Kobe 657, Japan*
[b] *Department of Management and Information, Osaka Junior College, Osaka 587, Japan*
[c] *Yunnan Bureau of Geology and Mineral Resources, Kunmin, China*

Received June 22, 1992; revision accepted March 2, 1993

ABSTRACT

More than 150 samples were collected at 23 sites from the Lower Cretaceous Jingxing Formation near the city of Yongping (25.5°N, 99.5°E), which is located on the west side of the Red River fault. Sixteen sites have characteristic directions with a high-temperature component above 500°C. The high-temperature component of magnetization from twelve sites is of pre-fold origin, and reveals clockwise deflection in declination ($D = 42.0°$, $I = 51.1°$ and $\alpha_{95} = 15.7°$). The easterly declination of more than 40° at Yongping is consistent with the Cretaceous paleomagnetic direction ($D = 45°$) of Chuxiong (25°N, 101.5°E), which is located on the east side of the Red River fault. Both areas were subjected to about 25° clockwise rotation with respect to the eastern part of the Yangtze block since Cretaceous time. We conclude that the Red River fault was rotated through $25° \pm 16°$ with respect to the eastern part of the Yangtze block, associated with rotation of the Yongping and Chuxiong areas. Removal of the rotation from the Red River fault indicates that the latter and the Jinsha suture formed a straight line along N55°W–S55°E within the Asian continent from Vietnam to the Tibetan Plateau in the present coordinate system during the Cretaceous. The linear feature of the Red River fault and Jinsha suture was probably deformed to the present curved shape as a result of collision of the Indian continent with the Asian continent.

1. Introduction

A unusual kinked shape is observed in the geomorphology of eastern Tibet and Yunnan (Fig. 1). Three large rivers, the Saluween, the Mekong and the Yangtze, flow eastward on the Tibetan Plateau. These rivers change their directions of flow to north–south around the Qamdo Diqu (31°N, 97°E), and concentrate in a narrow zone with a width of about 100 km in eastern Tibet. The Yangtze river again changes its direction to eastward in Central Yunnan. The Hengduan Shan mountain range curves between these three rivers.

This geomorphological kink is a reflection of the curvilinear shape of the terranes and sutures [1–3] (Fig. 1). The central part of the Tibetan Plateau is composed of three terranes (the Lhasa, the Qiangtang and the Bayan Kala) that are separated from each other by the east–west trending Banggong–Nujiang suture and the Jinsha suture (see Fig. 1). The Qiangtang and Lhasa terranes are bent and constricted in eastern Tibet, where the bending of the sutures forms the shape of a crank. The major rivers flow parallel to the shape of the sutures.

† Shoubu Funahara tragically disappeared in an avalanche on Mt. Meylixueshan in Yunnan Province on January 3, 1991. Correspondence to: Y. Otofuji, Department of Earth Sciences, Faculty of Science, Kobe University, Kobe 657, Japan

0012-821X/93/$06.00 © 1993 – Elsevier Science Publishers B.V. All rights reserved

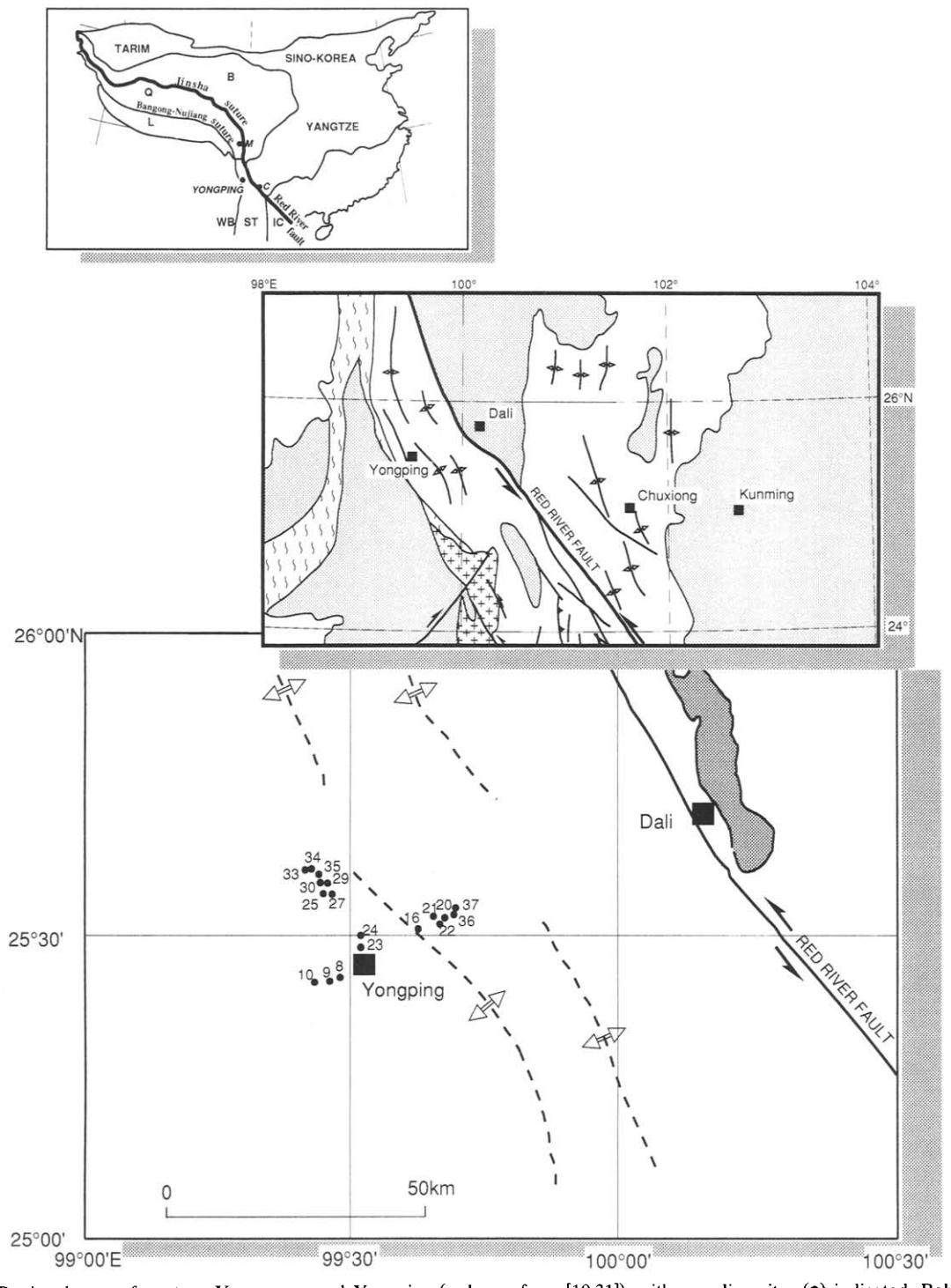


Fig. 1. Regional map of western Yunnan around Yongping (redrawn from [10,31]), with sampling sites (•) indicated. Bold line indicates the Red River fault. Major granites, metamorphic belt, and Tertiary anticlines are also shown. Sketch map of the tectonic blocks of southeastern part of the Asian continent is also indicated (top figure). *L* = Lhasa Terrane; *Q* = Qiangtang Terrane; *B* = Bayan Kala Terrane; *WB* = West Burman block; *ST* = Shan–Thai–Malay block; *IC* = Indochina block; *Y* = Yongping; *C* = Chuxiong; *M* = Markam.

There is a long-standing problem in the tectonic evolution of the shape of the sutures consisting in determining whether the crank shape is a nonrotational tectonic feature or whether the bend is product of suture rotation associated with terrane deformation. Paleomagnetic studies, however, have played an important role in detecting the rotation and deformation of the terrane blocks.

The Jinsha suture and Red River fault display very well the characteristic crank shape (Fig. 1), and we have therefore focused our paleomagnetic study on the rotation of the Red River fault in western Yunnan. Seeing as clockwise deflection of more than 45° has been observed in the Cretaceous paleomagnetic direction in the Chuxiong area on the east side of the Red River fault [4] we expect clockwise paleomagnetic rotation in the Yongping area, located on the west side of the Red River fault (Fig. 1).

Our expedition, therefore, in 1989, was to the Yongping area of central Yunnan, where we collected more than 150 samples from the Cretaceous red beds.

2. Geological setting and sampling

The Yongping area of western Yunnan forms the northern part of the Shan–Thai–Malay block, which is bounded to the northeast by the Yangtze block, to the east by the Indochina block and to the west by the West Burman block [5]. The Shan–Thai–Malay and Indochina blocks have formed the stable southern margin of Asia since at least the Triassic [6,7]. The Red River fault is the boundary between the Shan–Thai–Malay–Indochina and the Yangtze blocks and connects with the Jinsha suture in eastern Tibet. The Red River fault–Jinsha suture bends at about 23.4°N, 102.0°E: the strike of the Jinsha suture–Red River fault varies between N10°W and N35°W to the north of this point; the strike of the Red River fault fluctuates between N55°W and N70°W to the south of 23.4°N, 102.0°E.

Mesozoic rocks occur in the eastern part of the Shan–Thai–Malay block along the Red River fault in western Yunnan [8,9]; the Yongping area is located in the Lanping–Simao region. The Upper Jurassic to Cretaceous formations are terrestrial deposits consisting of purple-red, greyish green and yellowish sandstone–mudstones. The Upper Jurassic Bazhulu Formation disconformably underlies the Lower Cretaceous Jingxing Formation; the Jingxing Formation is conformably overlain by the Upper Cretaceous Mangang and Mankuanhe Formations. The Jingxing Formation occurs extensively throughout this region, and is assigned to the Lower Cretaceous on the basis of the fossil assemblages of *Peregrinoncocha* of *Trigonioides* (T.N.P.) and of *Cypride*, *Monosulcocypris*, *Damonella*, *Orthestheria* and *Lanpingosuchus magnus* etc. [8]. The folding took place in the Paleogene [10,11], resulting in north–south trending fold axes in the Mesozoic system of Yunnan Province.

We have collected samples at 23 sites from the Lower Cretaceous Jingxing Formation (Fig. 1). At the sampling sites this formation consists of massive red sandstones and red mudstones. No internal deformation was observed in the rocks. The bedding attitude at the sampling sites varies from 16° to 70°. Hand samples were oriented with a magnetic compass. At each site, we collected approximately seven independently oriented samples distributed over a distance ranging up to 50 m. The present declination value at each sampling point was evaluated from calculation using the International Geomagnetic Reference Frame of 1989 [12].

3. Paleomagnetism

3.1 Paleomagnetic measurements

In the laboratory, individual specimens 25 mm in diameter and 25 mm long were prepared from five samples from each site. Natural remanent magnetization (NRMs) was measured with an ScT two-axis cryogenic magnetometer. All specimens were subjected to stepwise thermal demagnetization up to 710°C. The results for each specimen were plotted on orthogonal vector diagrams [13] to assess the directional stability as well as on equal-area projections. Principal component analysis [14] was used to estimate the direction presented by the collinear array of five or more data points defining the component of magnetization.

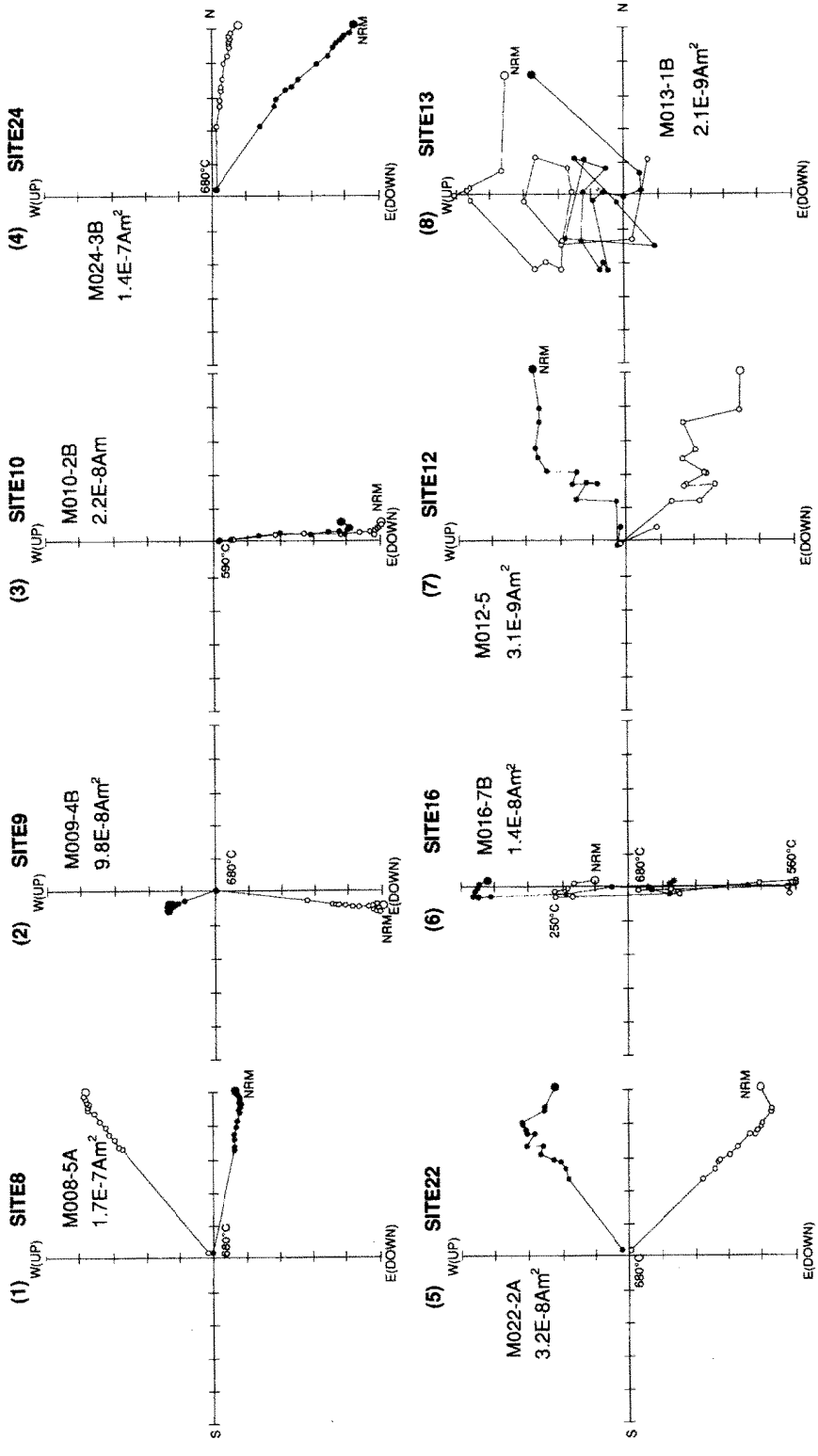


Fig. 2 Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens in the three categories A, B and C (see text). (1–5) Category A. (6) Category B. (7 and 8) Category C. Open symbols show the magnetic vector projection on to the vertical plane, solid symbols on to the horizontal plane. Numbers by symbols represent temperature (°C). Tectonic tilt correction has not been applied.

3.2 *Demagnetization*

Progressive thermal demagnetization of samples separated the sampling sites into three categories (Fig. 2 and Table 1): (A) single-component magnetization with high unblocking temperature, (B) multi-component magnetization, and (C) very variable behaviour within one site.

Category A: All samples at fifteen sites exhibited single-component magnetization (Fig. 2 (1–5)) and their stable NRM directions clustered. Unblocking temperature ranges around 700°C, indicating that this component is carried by hematite. Some samples had a viscous remanent magnetization acquired during storage in the laboratory, and this was removed below 100°C.

Category B: One site (site 16) exhibited three components of magnetization (Fig. 2 (6)). Thermal cleaning effectively resolved a discrete high-temperature component from all five samples between 450°C and 680°C; hematite was presumably the magnetic carrier. The medium-temperature component was removed from two samples at 500°C; its direction is antipodal to that of the high-temperature component. The low-temperature component magnetization was removed from four samples in the initial stage of demagnetization by 250°C and its direction seems to be paral-

TABLE 1

Paleomagnetic results for the Cretaceous rocks from Yongping. N is the number of field-oriented samples. Demagnetization: A = single-component magnetization; B = multi-component magnetization; C = very variable behaviour within one site; HT = high-temperature component; MT = medium-temperature component; LT = component destroyed at low temperature. Temperature range used to calculate the characteristic direction by principal component analysis is shown in the column headed °C. D and I are declination and inclination, respectively; k is the Fisherian precision parameter; α_{95} is the radius of cone of 95% confidence within site and overall mean. Data marked with * and # are not used for calculation of the mean direction for the Jingxing Formation

Site	Demagnetization Component (°C)			N	In situ				Tilt corrected				Locality	
					D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)	Lat. (°N)	Long. (°E)
Jingxing Formation (Lower Cretaceous)														
4	A	HT	100–680	3	63.9	23.0	157.6	9.9	21.4	34.0	157.6	9.9	25.46	99.52
8	A	HT	150–700	5	14.8	−34.8	156.6	6.1	−51.3	−68.5	156.6	6.1	25.43	99.50
9	A	HT	200–690	5	−156.7	72.8	82.6	8.5	121.3	46.3	82.6	8.5	25.43	99.47
10	A	HT	100–620	5	86.1	50.9	363.0	4.0	63.8	39.9	363.0	4.0	25.42	99.47
12	C												25.48	99.58
13	C												25.48	99.58
16	B	HT	460–680	5	55.2	76.7	103.9	8.6	72.0	53.0	103.9	8.6	25.50	99.61
		MT	250–500	2	−97.0	−48.5	–	–	−99.5	−25.8	–	–		
		LT	0–250	4	53.9	51.7	13.2	26.2	63.1	35.3	13.2	26.2		
20	A	HT	200–700	4	114.3	−70.5	23.6	19.3	169.2	−49.3	23.6	19.3	25.53	99.68
21	A	HT	100–710	5	−161.9	−72.0	33.3	13.5	−139.4	−44.5	33.3	13.5	25.53	99.68
22 #	A	HT	100–680	5	−28.0	47.9	21.1	17.1	−44.4	34.1	21.1	17.1	25.52	99.68
23	A	HT	100–700	5	78.8	17.0	95.8	7.9	55.1	53.5	95.8	7.9	25.48	99.53
24	A	HT	0–680	5	36.4	4.3	259.6	4.8	20.6	36.3	259.6	4.8	25.50	99.53
25	C												25.56	99.48
27	A	HT	100–680	5	46.1	3.1	46.3	11.4	50.4	49.4	46.3	11.4	25.57	99.48
28	A	HT	150–690	5	21.8	−27.2	17.9	18.6	18.7	37.7	17.9	18.6	25.57	99.47
29 #	A	HT	100–690	5	41.6	49.5	47.2	11.3	35.8	15.6	47.2	11.3	25.56	99.47
30 *	B	HT	500–700	3	−177.8	5.0	27.8	23.8	176.7	21.5	27.8	23.8	25.59	99.47
		LT	0–350	1	30.4	42.6	–	–	36.1	16.3	–	–		
33 *	B	HT	500–700	4	61.6	−16.5	1.1	180.0	65.8	43.9	1.1	180.0	25.61	99.45
		LT	100–560	5	151.5	40.5	1.5	103.0	−165.0	40.9	1.5	103.0		
34 #	A	HT	0–700	5	9.1	36.9	15.6	20.0	−43.9	69.9	15.6	20.0	25.60	99.41
35	A	HT	0–700	5	3.5	34.8	135.0	6.6	12.7	28.7	135.0	6.6	25.60	99.45
36 *	C	HT	300–680	5	−89.9	16.6	5.3	36.8	−80.8	48.3	5.3	36.8	25.53	99.68
37 #	A	HT	0–670	5	8.7	39.4	162.3	6.0	−19.3	56.0	162.3	6.0	25.53	99.68
39	C												25.55	99.70
mean		HT		12	48.9	50.8	2.7	33.4	42.0	51.1	8.7	15.7	25.50	99.50

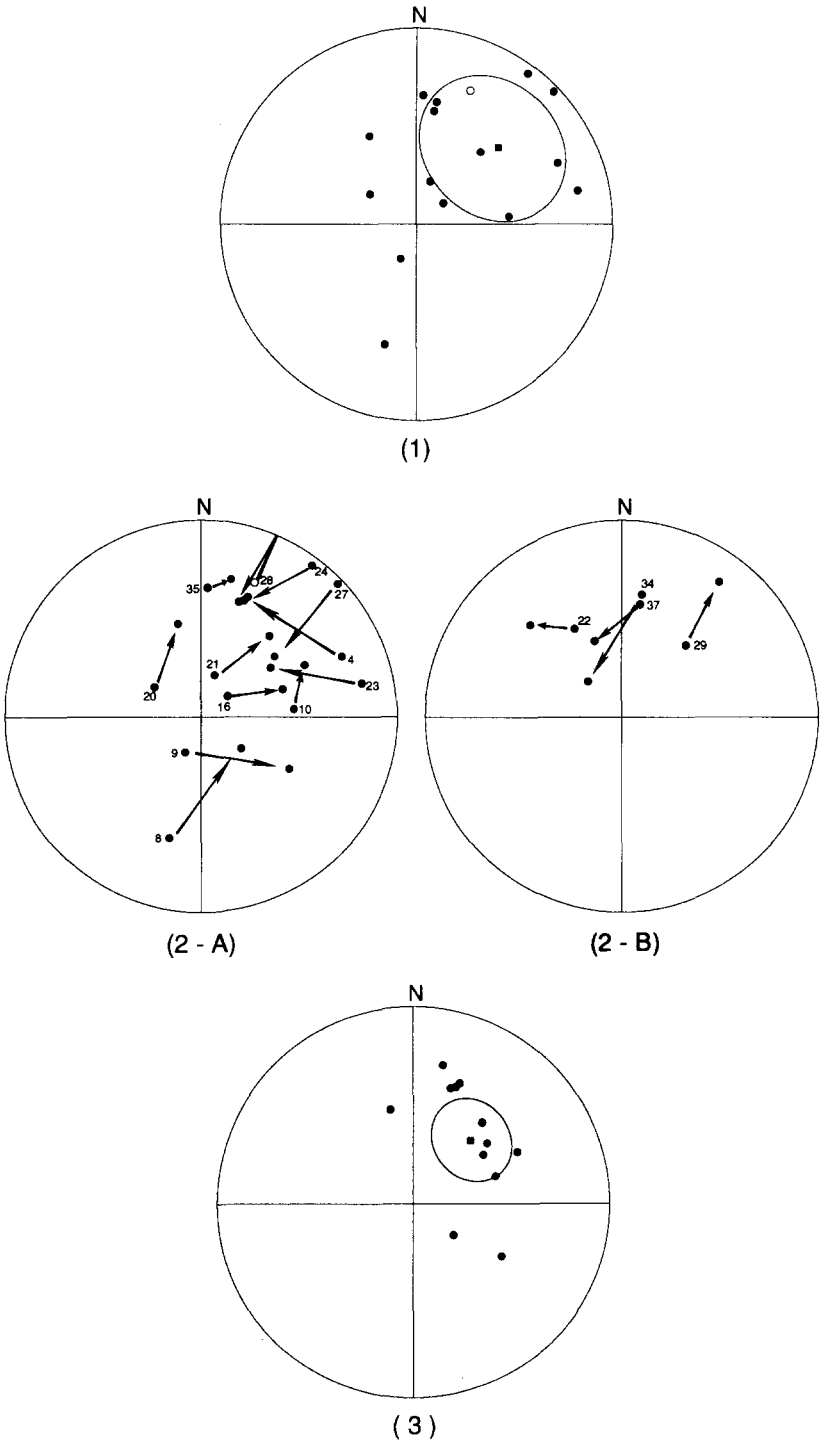


Fig. 3. Equal-area projections of the site mean directions (● and ○) for the Jinxing formation before and after tilt correction. All the reversed directions have been inverted through the origin. (1) The site mean directions of sixteen sites before tilt correction. Three data denoted by * in Table 1 are not shown in this figure. (2-A) Group A: Paleomagnetic directions from twelve sites in sixteen. They converge after correction for tilt. (2-B) Group B: Paleomagnetic directions of four sites in sixteen sites denoted by # in Table 1. They disperse after correction for tilt. (3) The site mean directions of twelve sites whose characteristic directions converge after correction for tilt. The solid symbols refer to the lower hemisphere, the open symbols to the upper hemisphere. Mean direction for each group is drawn as a square with a circle of 95% confidence in (1) and (3).

lel to the high-temperature component (Table 1). The NRM directions of the high-temperature component were well grouped.

Although two sites (site 30 and 33) exhibited two distinct magnetization components, their characteristic directions were dispersed within site.

Category C: Samples from site 36 exhibited either single- or two-component magnetization. Both the single-component and the high-temperature component in the two-component magnetization showed similar directions, with unblocking temperatures around 700°C.

Samples from four sites (12, 13, 25 and 39) had weak magnetization, and revealed completely erratic demagnetization paths after their intensity decreased to less than 10^{-4} A/m (Fig. 2 (7 and 8)). No characteristic directions could be isolated from these sites.

3.3 Tilt correction

The characteristic direction could be isolated at sixteen sites (Table 1). Magnetization consisting in a single component or high-temperature component with an unblocking temperature around 700°C was recognized to be the characteristic magnetization. The site mean direction of the characteristic directions of each site was calculated by assigning a unit weight to each sample. We arbitrarily discarded results from any site for which the radius of cone of 95% confidence exceeded 20°.

The site mean directions for the sixteen sites are spread mainly in the first quadrant, with two reversed results *in situ* (Fig. 3 (1)). The mean direction of the sixteen site mean directions is $D = 34.8°$, $I = 50.8°$, with a large α_{95} of 24.5°.

The paleomagnetic results for the Jingxing Formation are divided into two groups (A and B) on the basis of the tilt correction. When the site mean directions are corrected for tilt, twelve sites (group A) show a convergence of their site mean directions toward a northeasterly and intermediate downward direction (Fig. 3 (2-A)). The tilt-corrected mean paleomagnetic direction of the twelve sites is $D = 42.0°$, $I = 51.1°$, with $\alpha_{95} = 15.7°$. The precision parameter k increases from 2.7 to 8.7 after tilt correction for the twelve sites.

The remaining four sites (22, 29, 34 and 37) form the other group (group B). Their site mean directions reveal a divergence from the uncorrected mean direction during tilt correction (Fig. 3 (2-B)). All four sites reveal normal polarity and their mean direction is similar to the present geomagnetic field ($D = 7.8°$, $I = 45.9°$) *in situ*. The precision parameter decreases from 16.1 to 5.0 after tilt correction. The rocks of this group probably acquired their magnetization after tilting occurred.

4. Discussion

4.1 Characteristic direction of the Jingxing Formation

Our paleomagnetic study indicates that the twelve group A sites acquired their stable magnetization before folding, whereas the remaining four sites (group B) acquired their magnetization after folding. The group A sites probably preserve the reliable paleomagnetic direction of the Lower Cretaceous Jingxing Formation. These twelve group A sites provide a positive fold test which is significant at 99% confidence level [15]. We recognize here that the directions of site 8, 20 and 21 are of reversed polarity because of their negative inclination value. The mean of the group A sites ($D = 42.0°$, $I = 51.1°$, with $\alpha_{95} = 15.7°$; Fig. 3(3)) probably represents the characteristic formation mean of the Jingxing Formation.

The dispersion of the declinations in the paleomagnetic directions of the group A sites appears to be larger than that of the inclinations (see Fig. 3 (3)). This is presumably due to the simple tilt correction for the strata, which dip quite steeply. Seeing as the dip at nine of the twelve sites is greater than 30°, insufficient accuracy in the tilt correction might induce the dispersion of declinations in the paleomagnetic direction on the equal-area projection. Fold axes may have undulated in the Yongping area, and this increased variation in the fold axes increases the dispersion of declination after simple tilt correction. Seeing as the error in tilt correction and variation of the plunge of the fold axes seem to be randomly distributed, the Fisherian mean of the group A sites is a reasonable characteristic direction of the Yongping area.

The demagnetization behaviour of the two

TABLE 2

Observed (obs) and expected (exp) paleomagnetic directions for the Cretaceous: Mean paleomagnetic direction for Cretaceous formations of Yongping, and Virtual Geomagnetic Pole position (VGP). D_{obs} and I_{obs} for Yongping are declination and inclination of the observed direction for the Jingxing Formation. (D_{obs} and I_{obs} on the Red River fault are calculated on the basis of the observed data from the Yongping and Chuxiong areas.) α_{95} is the radius of cone of 95% confidence; ΔD is uncertainty of declination ($\Delta D = \sin^{-1}(\sin \alpha_{95}/\cos I)$). D_{exp} and I_{exp} are declination and inclination of the expected direction for Yongping converted from the VGP for each area, which is calculated from the paleomagnetic data in the studies cited

Area	N	Observed direction				V.G.P.			References
		D_{obs}	I_{obs}	α_{95}	ΔD_{obs}	Lat.	Long.	α_{95}	
Yongping (25.5°N, 99.5°E)	12	42.0°	51.1°	15.7°	25.5°	50.9°N	167.3°E	20.6°	This study
Chuxiong (25.0°N, 101.5°E)	21	44.6°	41.3°	10.7°	14.3°	49.2°N	182.0°E (178.0°W)	11.4°	[4]
Red river fault (25.3°N, 100.5°E)	33	43.7°	44.7°	8.7°	12.3°	50.0°N	177.0°E	10.0°	–
Area	N	Expected direction at Yongping				V.G.P.			References
		D_{exp}	I_{exp}	α_{95}	ΔD_{exp}	Lat.	Long.	α_{95}	
Whole China block	12	13.9°	39.6°	–	–	77.0°N	200.0°E	3.2°	[26]
Eastern part of the Yangtze block	5	19.1°	37.9°	7.9°	10.0°	72.1°N	198.6°E	7.2°	[21–25]
Markam (Qiangtan Terrane)	3	46.8°	48.6°	9.7°	14.8°	48.6°N	172.9°E	12.6°	[35,36]

groups is identical (the unblocking temperature is around 700°C) (see Fig. 2). Hematite carries the secondary, acquired magnetization of the group B sites (Fig. 2 (5)) and the primary component of the group A sites (Fig. 2 (1–4)). It is difficult to explain why hematite carries secondary magnetization at only four sites in the Jingxing Formation, but one possible explanation is an error in the sampling of the formation—seeing as the Upper Jurassic Bazhulu Formation outcrops in

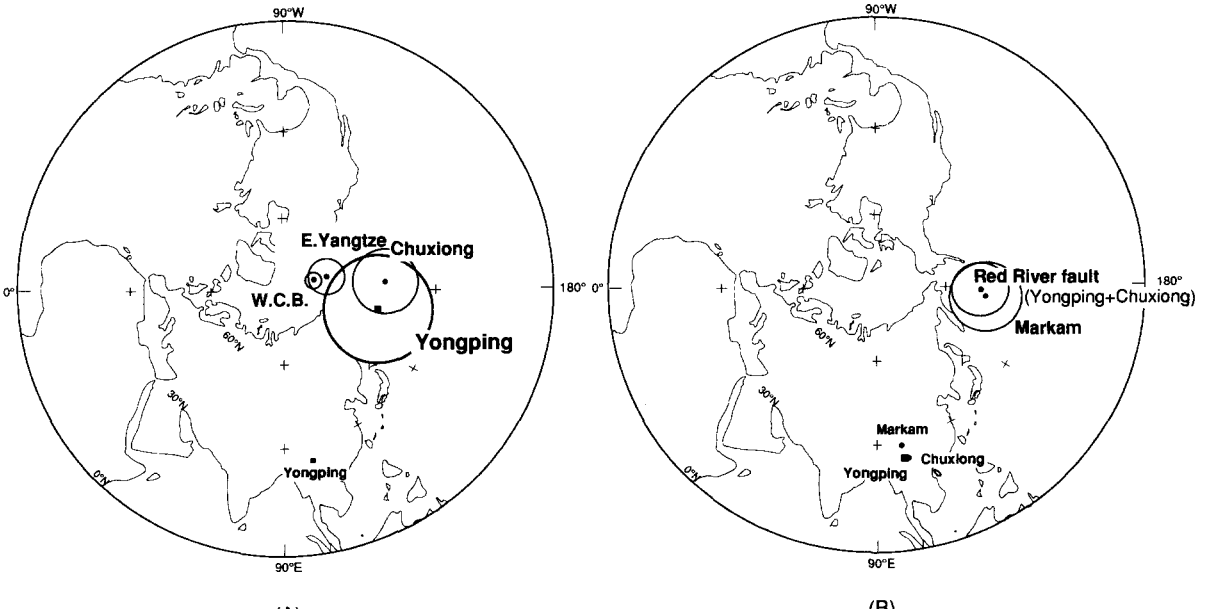


Fig. 4. Cretaceous paleomagnetic poles (with associated circles of 95% confidence) for the Yongping area and some parts of Asia. (A) Poles for the Yongping area and the Chuxiong area together with poles for the eastern Yangtze block and the whole-China block (*W.C.B*). (B) Poles for the Red River fault near the Yongping and Chuxiong areas and the Markam area in the Qiangtang Terrane. Pole and location of Yongping are shown by square. Polar stereographic projection.

the Yongping area, the four group B sites may actually be located in this formation. Further, recent paleomagnetic studies have suggested that the Jurassic and older formations experienced pervasive remagnetization over an extensive area of southern China [7,16–19]. We therefore exclude the paleomagnetic results of the secondarily magnetized group from the characteristic directions of the Jinxing Formation.

4.2 Easterly paleomagnetic direction of the Jingxing Formation

A large clockwise deflection in declination ($D = 42°$) appears in the paleomagnetic directions of the Lower Cretaceous rocks of the Yongping area. The reliability of the easterly direction is ascertained through the presence of normal and reversed polarities and through thermal demagnetization and the fold test. The reversed direction of site 21 is reasonably antipodal to the normal ones. Although site 8 and 9 are far removed from the mean direction, they are antipodal each other. The positive fold test demonstrates this easterly direction as a prefolding one. Because the folding in the study area probably took place in the Paleogene [10,20], the prefolding magnetization is probably the Cretaceous paleomagnetic direction.

The easterly direction of the Cretaceous has been observed in the Chuxiong area, which is on the east side of the Red River fault [4] (Fig. 1); the declination value in this area is 45°. The mean direction converted to the Virtual Geomagnetic Pole (VGP) for the Cretaceous of the Yongping area gives 50.9°N, 167.3°E, with $\alpha_{95} = 20.6°$ (Table 2), whereas the VGP position of the Chuxiong area is 49.2°N, 182.0°E (178.0°W), with $\alpha_{95} = 11.4°$. A comparison of the two VGP positions is made in Fig. 4. Although the polar distance between the two poles is fairly large (9.6°), two circles of 95% confidence overlap, indicating that the VGPs for the Yongping and Chuxiong areas are not significantly different. Yongping and Chuxiong (which are on either side of the Red River fault) preserve the same easterly paleomagnetic direction of the Cretaceous.

A large clockwise rotational motion is expected for the Yongping and Chuxiong areas with respect to the rest of China. The geomagnetic poles of the Yongping and Chuxiong areas are compared with the Cretaceous poles from the eastern part of the Yangtze block and the whole-China block in Fig 4. The VGP from the eastern part of the Yangtze block is located at 72°N, 199°E, with $\alpha_{95} = 7°$ [21–25]. The VGP of 90 Ma for the Eurasian continent (77°N, 200°E, with $\alpha_{95} = 3°$) [26] is recognized to be the Cretaceous VGP of the whole-China block. Although the Yongping pole has a fairly large circle of 95% confidence, within 95% confidence it is significantly different from the poles of the eastern part of the Yangtze block and the whole-China block. The poles from the Yongping and Chuxiong areas are situated to the east of the Cretaceous poles from the eastern part of the Yangtze block and whole-China block, clearly illustrating the clockwise rotation of these areas.

The Yongping area is rotated clockwise through 23° with respect to the eastern part of Yangtze and through 28° with respect to Eurasia since Cretaceous time. The amount of rotation R is given by $R = D_{obs} - D_{exp}$, where D_{obs} and D_{exp} are the observed and expected declination at Yongping. The expected declination value (D_{exp}) is calculated from the VGPs of the eastern part of Yangtze and 90 Ma Eurasia (Table 2). Seeing as the Chuxiong area was rotated through 31° with respect to Eurasia since the Cretaceous [4], more than 25° is assigned to the clockwise rotation with respect to Eurasia for the two areas separated by the Red River fault.

4.3 Tectonic rotation of the Red River fault

We shall now address the question of whether the Yongping and Chuxiong areas rotated individually as independent blocks or whether these areas behaved together as a single block. Right-lateral strike-slip faulting is capable of resulting in individual clockwise rotations for two blocks along a fault [27–30]. The local deformation associated with strike-slip on the Red River fault, however, did not cause the clockwise rotations of the Yongping and Chuxiong areas because in the Tertiary the large-scale movement of the Red River fault was left-lateral [10], and the only right-lateral strike-slip displacement known on this fault was small-scale Quaternary movement [10,31]. Further, in regard to the latter point,

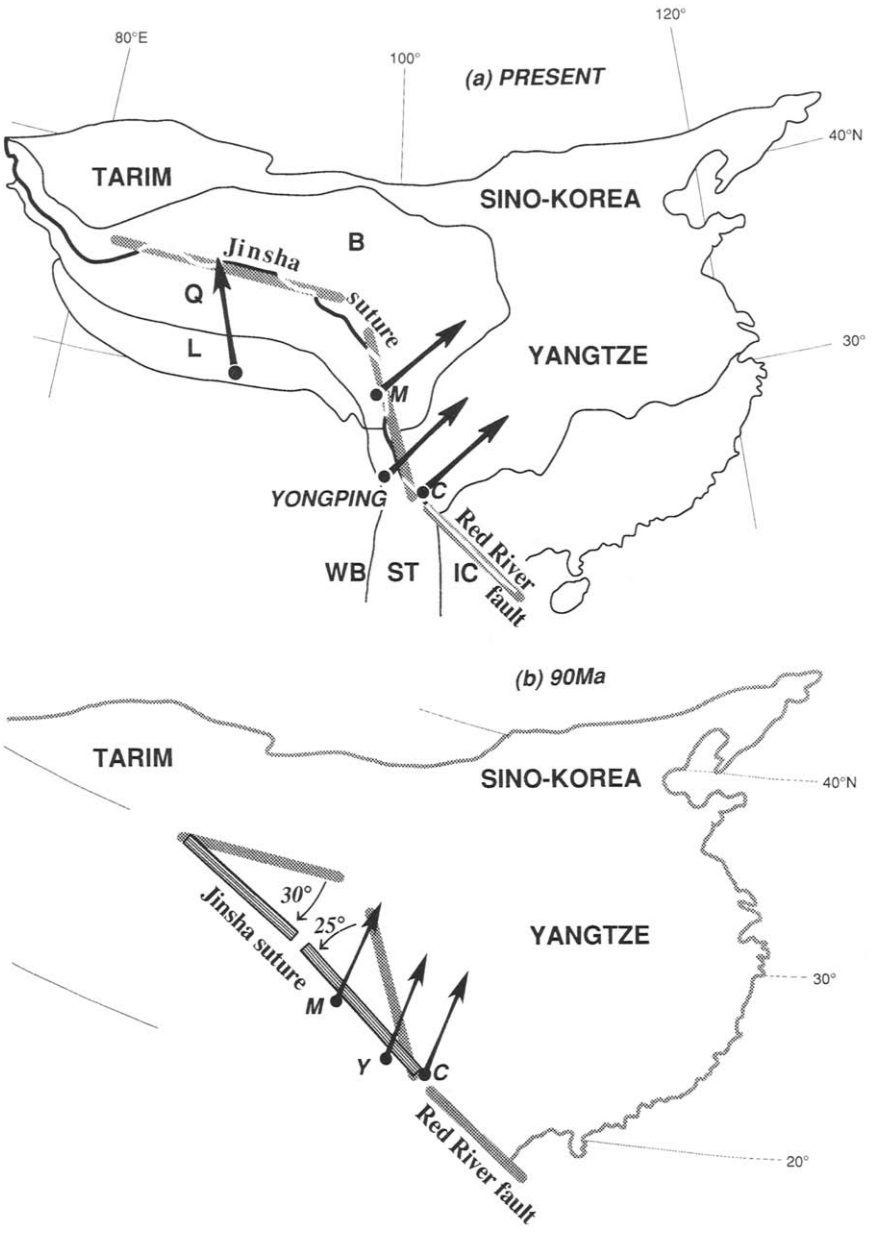


Fig. 5. Arrangement of the Red River fault–Jinsha suture at the present (a) and at 90 Ma in the Cretaceous (b). (a) Present: The Red River fault–Jinsha suture is curvilinear in shape and may be divided into three parts: a northern part with a strike of N80°W on the Tibetan Plateau, a central part with the strike of N15°W in the Three Rivers region, and a southern part along the Red River with a strike of N55°W. The solid arrow marks the observed declination of the Cretaceous rocks in Yongping (*Y*), Chuxiong (*C*), Markam (*M*) and Lhasa (*L*). *L* = Lhasa Terrane; *Q* = Qiangtang Terrane; *B* = Bayan Kala Terrane; *WB* = West Burman block; *ST* = Shan–Thai–Malay block; *IC* = Indochina block. (b) 90 Ma: Ancestral shape of the Red River fault–Jinsha suture is reconstructed. Chuxiong and the southern part of the Red River fault–Jinsha suture are arranged relative to the whole-China block. The northern part is restored by clockwise rotation through 30°, and the central part is rotated counterclockwise through 25°. Ancestral strikes of the northern and central parts were N60°W and N40°W, respectively, in the present coordinate system in the Cretaceous. The ancestral shape is thus expected to have been linear in the Cretaceous. The Cretaceous paleomagnetic directions in Yongping, Chuxiong and Markam become nearly parallel to the Cretaceous paleomeridian associated with the reconstruction of the Red River fault and the Jinsha suture. The paleolatitude in the Cretaceous is calculated from the pole at 90 Ma (77°N, 200°E) [26].

seeing as Yongping and Chuxiong are situated more than 30 km from the Red River fault, they escaped any deformation caused by this small-scale right-lateral faulting. To conclude, strike-slip movement on the Red River fault can not account for the individual clockwise rotations of the Yongping and Chuxiong areas.

An alternative explanation is a single rotation of a fairly large block which consists of the Yongping and Chuxiong areas and the Red River fault. The geomorphology of a curved fault can sometimes preserve information about crustal rotation associated with deformation within a block [e.g. 32–34]. As stated earlier, the Red River fault is bent at 23.4°N, 102.0°E, just south of Yongping and Chuxiong. The strike of the Red River fault north of this point varies between N65°W and S65°E, whereas the strike to the south is between N35°W and S35°E [10]. The northern part thus appears to be bent clockwise with respect to the southern part by 25°, which is consistent with clockwise rotation of the Yongping and Chuxiong areas with respect to the eastern part of Yangtze block. We conclude that the area north of the bend at 23.4°N, 102.0°E along the Red River fault formed a single tectonic block and underwent tectonic rotation. The Red River fault was consequently rotated about a vertical axis associated with the rotational motion of the single block including the Yongping and Chuxiong areas.

To evaluate the amount of rotation of the Red River fault, we assume that the rotation of the fault is represented by that of the Yongping and Chuxiong areas since the Cretaceous. We calculate the amount of rotation of a block including the Yongping and Chuxiong areas on the basis of the Cretaceous data from the two areas; the observed value from the two areas is $D = 43.7°$, $I = 44.7°$ (with $\alpha_{95} = 8.7°$) at the reference point 25.3°N, 100.5°E. The amount of rotation (R) and its uncertainty (ΔR) is $25° \pm 16°$ with respect to the eastern part of Yangtze. The uncertainty ΔR is given as $\Delta R = [(\Delta D_{obs})^2 + (\Delta D_{exp})^2]^{1/2}$, where ΔD_{obs} and ΔD_{exp} are the uncertainties at the 95% confidence level ($\Delta D = \sin^{-1}$ (sin $\alpha_{95}/\cos I$).

A large clockwise deflection in declination ($D = 48°$) has also been reported from Cretaceous rocks of the Markam area in the Qiangtang Terrane [35,36]. The Markam area is 500 km north of Yongping and Chuxiong, and is located on the western side of the Jinsha suture. The Cretaceous pole from Markam (49°N, 173°E, $\alpha_{95} = 13°$) is not significantly different from the pole from Yongping and Chuxiong (Fig. 4), and the good agreement of these Cretaceous poles indicates that little relative rotation or translation occurred between the Yongping, Chuxiong and Markam areas. This implies that a large area covering Chuxiong to Markam across the Red River fault and Jinsha suture experienced the same tectonic rotation. The Red River fault and Jinsha suture, which ranges all the way from the Yongping and Chuxiong areas to the Markam area, was subjected to clockwise rotation of more than 25° together with this large area.

Significant latitudinal change does not, however, accompany this clockwise rotation. We compare the observed declination values at Yongping and at the Red River fault with the inclination values expected from the eastern part of the Yangtze (see Table 2). The differences in inclinations ($I_{obs} - I_{exp}$) and their uncertainty ($[(\alpha_{95\ obs})^2 + (\alpha_{95\ exp})^2]^{1/2}$) are $13.2° \pm 17.6°$ for Yongping and $6.8° \pm 11.8°$ for the Red River fault; this is not statistically significant. The Red River fault and the Jinsha suture experienced clockwise rotation with insignificant north–south translation.

4.4 The ancestral shape of the Red River fault and Jinsha suture

The formation of the crank shape of the Red River fault and Jinsha suture is thus explained by the rotation of the fault or suture. We shall estimate the shape of the suture and fault before deformation by using the paleomagnetic data. The Jinsha suture and the Red River fault are roughly divided into three parts (Fig. 5), the northern part running along N80°W on the Tibetan Plateau, the central part running along N15°W in the Three Rivers region, and the southern part running along the Red River with a strike of N55°W. We assume here that the southern part has been in its present position since the Cretaceous.

We recognize that the amount of rotation of the central part corresponds to that of the Red River fault in central Yunnan. Seeing as the Red River fault at 25.3°N, 100.5°E has been rotated

clockwise through 25° ± 16° with respect to the eastern part of Yangtze since the Cretaceous, the central part may be restored by a counterclockwise rotation of 25° (Fig. 5). The original strike of the central part of the Red River fault–Jinsha suture is estimated to be N40°W in the present coordinate system.

Although no Cretaceous paleomagnetic data have been reported from the vicinity of the northern part of the Jinsha suture, counterclockwise rotation of the Tibetan Plateau area has been suggested from the Jurassic paleomagnetic direction of the Qiangtang Terrane [5] and the Cretaceous data of the Lhasa Terrane [35,37]. We use 30° of counterclockwise rotation of the Lhasa Terrane with respect to the eastern part of Yangtze for unbending the northern part of the Jinsha suture [35]. Reconstruction shows that the original strike of the northern part is restored at N60°W in the present coordinate system.

Removal of the rotation from the northern and central parts of the Jinsha suture and the Red River fault creates an approximately linear trend from Vietnam to the Tibetan Plateau prior to the Cretaceous. The reconstructed strike varies between N40°W and N60°W. Such long straight terrane boundaries have already been observed in the Paleozoic accretionary belts of the Appalachian orogen [38], and thus this linear feature is a realistic estimate for the ancestral shape of the Jinsha suture–Red River fault in the Cretaceous.

The trend of the Jinsha suture is oblique rather than parallel to the present and Cretaceous latitude lines. Because the Cretaceous VGP positions of the Markam, Chuxiong and Yongping areas are similar, the three areas have preserved their relative latitude relationships since the Cretaceous (although Markam could have been situated at a higher latitude than Yongping and Chuxiong in the Cretaceous).

We conclude that the ancestral Jinsha suture and Red River fault were linear and oblique with strikes of N55°W in the present coordinate system during the Cretaceous.

Geophysical and geological studies indicate that the Indian continent collided with the Asian continent at about 50 Ma [e.g. 39–45]. This collision has produced significant crustal deformation within the Asian continent [46–51]. One manifestation of the deformation was probably the change in geometry of the Red River fault and the Jinsha suture from a linear feature to the crank shape. Paleomagnetic study has unveiled the tectonic bending in the Red River fault and Jinsha suture caused by the collision event.

5. Conclusion

(1) Since the Cretaceous the Yongping area has been rotated clockwise through 23° with respect to the eastern part of Yangtze.

(2) A large area covering Yongping, Chuxiong and Markam was subjected to a similar rotational motion of about 25° within the Asian continent.

(3) The Red River fault was rotated through 25° ± 16°, accompanied by the rotation of a large area encompassing Yongping, Chuxiong and Markam.

(4) The bend of the Red River fault at 23.4°N, 102.0°E is a geomorphological remnant of the clockwise rotation of the fault.

(5) The Jinsha suture and the Red River fault had an approximately linear trend stretching from the Tibetan Plateau to Vietnam during the Cretaceous. Its strike was about N55°W in the present coordinate system.

Acknowledgements

Samples were collected in 1989 during the Japan–China Joint Friendship Expedition to Mt. Meylixueshan organized by the Academic Alpine Club of Kyoto, the Yunnan Branch of the All-China Sports Federation, and the Chinese Mountaineering Association. The particularly efficient field logistics were set up by the Bureau of Geology and Mineral Resources of Yunnan Province. We are grateful to Nobuo Ishihama and Xiao-Chun Ti for sampling in the field. Yukiko Tanaka, Chris Brick and Steve D. Sheriff helped greatly in improving the English.

References

1 C.Y. Li, Q. Wang, X.Y. Liu and Y.Q. Tang, Explanatory notes to the tectonic map of Asia, 49 pp., Cartogr. Publ. House China, Beijing, 1982.

2 Zh.M. Zhang, J.G. Liou and R.G. Coleman, An outline of plate tectonics of China, Geol. Soc. Am. Bull. 95., 295–312, 1984.

3 J.L. Lin and M. Fuller, Paleomagnetic constraints on relative motions of the Asian blocks and terranes, in: Terrane Analysis of China and the Pacific Rim, Circ.-Pac. Counc. Energy Miner. Resour. Earth Sci. Ser. 13, 195–210, 1990.
4 S. Funahara, N. Nishiwaki, M. Miki, F. Murata, Y.-i. Otofuji and Y.Z. Wang, Paleomagnetic study of Cretaceous rocks from the Yangtze block, central Yunnan, China: implications for the India–Asia collision, Earth Planet. Sci. Lett. 113, 77–91, 1992.
5 J.L. Lin and D.R. Watts, Palaeomagnetic results from the Tibetan Plateau, Philos. Trans. R. Soc. London A327, 239–262, 1988.
6 E. Buffetaut, The paleo-bio-geographical significance of the Mesozoic continental vertebrates from South East Asia, Mem. Soc. Geo. Fr. 147, 37–42, 1984.
7 C. Yan and V. Courtillot, Widespread Cenozoic(?) remagnetization in Thailand and its implications for the India–Asia collision, Earth Planet. Sci. Lett. 93, 113–122, 1989.
8 Y.C. Hao et al., The Cretaceous System of China, 301 pp., Geol. Publ. House, Beijing, 1986.
9 Bureau of Geology and Mineral Resources of Yunnan Province, Regional Geology of Yunnan Province (Geological Memoirs, Ser. 1, No. 21), 727 pp., Geol. Publ. House, Beijing, 1990.
10 P. Tapponnier, R. Lacassin, P.H. Leloup, U. Schärer, Z. Dalai, H.W. Wu, X.H. Liu, S.C. Ji, L.S. Zhang and J.Y. Zhong, The Ailao Shan/Red River metamorphic belt: Tertiary left-lateral shear between Indochina and South China, Nature 343, 431–437, 1990.
11 U. Schärer, P. Tapponnier, R. Lacassin, P.H. Leloup, Z. Dalai and J. Shaocheng, Intraplate tectonics in Asia: a precise age for large-scale Miocene movement along the Ailao Shan–Red River shear zone, China, Earth Planet. Sci. Lett. 97, 65–77, 1990.
12 IAGA Division I Working Group 1, International geomagnetic reference field revision 1987, J. Geomag. Geoelectr. 39, 773–779, 1987.
13 J.D.A. Zijderveld, A.C. demagnetization of rocks: Analysis of results, in: Methods in Paleomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds., pp. 254–286, Elsevier, Amsterdam, 1967.
14 J.L. Kirschvink, The least-squares line and plane and the analysis of palaeomagnetic data, Geophys. J.R. Astron. Soc. 62, 699–718, 1980.
15 M.W. McElhinny, Statistic significance of the fold test in palaeomagnetism, Geophys. J.R. Astron. Soc. 8, 338–340, 1964.
16 D.V. Kent, X.S. Zeng, W.Y. Zhang and N.D. Opdyke, Widespread late Mesozoic to Recent remagnetization of Paleozoic and lower Triassic sedimentary rocks from South China, Tectonophysics 139, 133–143, 1987.
17 Y.J. Zhai, M.K. Seguin, Y.X. Zhou, J. Dong and Y.X. Zheng, New paleomagnetic data from the Huanan block, China, and Cretaceous tectonics in eastern China, Phys. Earth Planet. Inter. 73, 163–188, 1992.
18 J.P. Dobson and F. Heller, Remagnetization in southeast China and the collision and suturing of the Huanan and Yangtze Blocks, Earth Planet. Sci. Lett. 111, 11–21, 1992.
19 Z. Wang and R. Van der Voo, Pervasive remagnetization of Paleozoic rocks acquired at the time of Mesozoic folding in the South China Block, J. Geophys. Res., in press, 1992.
20 Academia Sinica, Geologic map of Yunnan, scale 1:1,000,000, Science Press, Beijing, 1981.
21 J.L. Lin, M. Fuller and W.Y. Zhang, Preliminary Phanerozoic polar wander paths for the North and South China blocks, Nature 313, 444–449, 1985.
22 D.V. Kent, G. Xu, K. Huang, W.Y. Zhang and N.D. Opdyke, Paleomagnetism of upper Cretaceous rocks from South China, Earth Planet. Sci. Lett. 79, 179–184, 1986.
23 Y. Otofuji, K.H. Kim, H. Inokuchi, H. Morinaga, F. Murata, H. Katao and K. Yaskawa, A paleomagnetic reconnaissance of Permian to Cretaceous sedimentary rocks in southern part of Korean Peninsula, J. Geomag. Geoelectr. 38, 387–402, 1986.
24 G.D. Lee, J. Besse, V. Courtillot and R. Montigny, Eastern Asia in the Cretaceous: New paleomagnetic data from south Korea and a new look at Chinese and Japanese data, J. Geophys. Res. 92, 3580–3596, 1987.
25 L.S. Chan, Paleomagnetism of Late Mesozoic granitic intrusions in Hong Kong: implications for upper Cretaceous reference pole of South China, J. Geophys. Res. 96, 327–335, 1991.
26 J. Besse and V. Courtillot, Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian Plates, and true polar wander since 200 Ma, J. Geophys. Res. 96, 4029–4050, 1991.
27 R. Freund, Rotation of strike slip faults in Sistan, southeast Iran, J. Geol. 78, 188–200, 1970.
28 M.E. Beck, Jr., Discordant paleomagnetic pole positions as evidence of regional shear in the Western Cordillera of North America, Am. J. Sci. 276, 694–712, 1976.
29 W.D. MacDonald, Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, J. Geophys. Res. 85, 3659–3669, 1980.
30 A. Nur, H. Ron and O. Scotti, Fault mechanics and the kinematics of block rotations, Geology 14, 746–749, 1986.
31 C.R. Allen, A.R. Gillespie, Han Yuan, K.E. Sieh, Zhang Buchun and Zhu Chengnan, Red River and associated faults, Yunnan Province, China: Quaternary geology, slip rates, and seismic hazard, Geol. Soc. Am. Bull. 95, 686–700, 1984.
32 R. Van Der Voo and J.E.T. Channell, Paleomagnetism in orogenic belts, Rev. Geophys. Space Phys. 18, 455–481, 1980.
33 S. Eldredge, V. Bachtadse and R. Van Der Voo, Paleomagnetism and the orocline hypothesis, Tectonophysics 119, 153–179, 1985.
34 Y. Itoh, Differential rotation of the eastern part of southwest Japan inferred from paleomagnetism of Cretaceous and Neogene rocks, J. Geophys. Res. 93, 3401–3411, 1988.
35 Y. Otofuji, Y. Inoue, S. Funahara, F. Murata and X. Zheng, Palaeomagnetic study of eastern Tibet—deformation of the Three Rivers region, Geophys. J. Int. 103, 85–94, 1990.
36 K.N. Huang, N.D. Opdyke, J.G. Li and X.J. Peng, Paleomagnetism of Cretaceous rocks from Eastern Qiangtang Terrane of Tibet, J. Geophys. Res. 97, 1789–1799, 1992.
37 J. Achache, V. Courtillot and Z.Y. Xiu, Paleogeographic and tectonic evolution of southern Tibet since middle

Cretaceous time: new paleomagnetic data and synthesis, J. Geophys. Res. 89, 10311–10339, 1984.
38 H. Williams and R.D. Hatcher, Jr., Appalachian suspect terranes, Geol. Soc. Am. Mem. 158, 33–54, 1983.
39 B.D. Johnson, C. McA. Powell and J.J. Veevers, Spreading history of the eastern Indian Ocean and Giant India's northward flight from Antarctica and Australia, Geol. Soc. Am. Bull. 87, 1560–1566, 1976.
40 P. Patriat and J. Achache, India–Eurasia collision chronology has implications for crustal shortening and driving mechanism of plates, Nature 311, 615–621, 1984.
41 C.J. Allegre et al., Structure and evolution of the Himalaya–Tibet orogenic belt, Nature 307, 17–22, 1984.
42 U. Schärer, R.H. Xu and C.J. Allegre, U-Pb geochronology of Gangdese (Transhimalaya) plutonism in the Lhasa–Xigaze region, Tibet, Earth Planet. Sci. Lett. 69, 311–320, 1984.
43 J. Besse and V. Courtillot, Paleogeographic maps of the continents bordering the Indian Ocean since the early Jurassic, J. Geophys. Res. 93, 11791–11808, 1988.
44 J.-Y. Royer and D.T. Sandwell, Evolution of the Eastern Indian Ocean since the late Cretaceous: Constraints from Geosat altimetry, J. Geophys. Res. 94, 13755–13782, 1989.
45 P. Tapponnier et al., The Tibetan side of the India–Eurasia collision, Nature 294, 405–410, 1981.
46 J.P. Vilotte, M. Daignieres and R. Madariaga, Numerical modeling of intraplate deformation: simple mechanical models of continental collision, J. Geophys. Res. 87, 10709–10728, 1982.
47 P. Tapponnier, G. Peltzer, A.Y. Le Dain, R. Armijo and P. Cobbold, Propagating extrusion tectonics in Asia: New insights from simple experiments with plasticine, Geology 10, 611–616, 1982.
48 P. England and D. McKenzie, A thin viscous sheet model for continental deformation, J. Geophys. Res. 70, 295–321, 1982.
49 G. Houseman and P. England, Finite strain calculations of continental deformation 1. Methods and general results for convergent zones, J. Geophys. Res. 91, 3651–3663, 1986.
50 G. Peltzer and P. Tapponnier, Formation and evolution of strike-slip faults, rifts, and basins during the India–Asia collision: an experimental approach, J. Geophys. Res. 93, 15085–15117, 1988.
51 P. Davy and P.R. Cobbold, Indentation tectonics in nature and experiment. 1. Experiments scaled for gravity, Bull. Geol. Inst. Upps. 14, 129–141, 1988.